

TERM PROJECT REVIEW

Digital Library Search System

Data Structures and Algorithms – 3 (25CS2103E)

Academic Year: 2026 – 2027

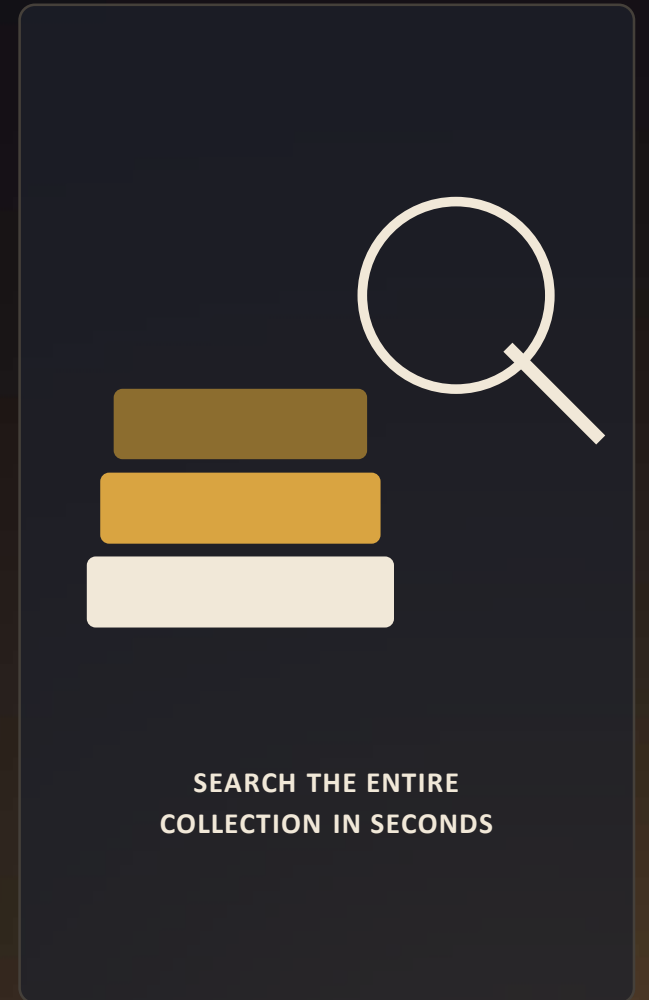
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PROJECT GUIDE

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ABSTRACT



- Libraries store a large number of books, journals, magazines, and research papers.
- Manual searching becomes inefficient as the collection size increases.
- The proposed system provides a centralized digital repository.
- Users can search using Title, Author, ISBN, Category, and Keywords.
- The project is implemented using Java and DSA concepts only.
- The focus is on efficient searching and organized retrieval of records.



Search by Title, Author, ISBN, Category or Keyword...

PROBLEM STATEMENT



THE CHALLENGE

- Large volume of digital library resources.
- Manual searching is slow and time-consuming.
- Users require multiple search options.
- Retrieval should be fast and accurate.
- The system should remain scalable as records increase.

JUSTIFICATION

- Efficient search algorithms reduce unnecessary comparisons.
- Hash-based lookup improves retrieval speed.
- Sorting enables organized presentation of records.
- The project demonstrates practical DSA application in information retrieval.

FAST

ACCURATE

SCALABLE

PROPOSED FUNCTIONALITIES



- Store and organize book records.
- Search by Title, Author, ISBN, Category, and Keywords.
- Fast retrieval using searching and hashing techniques.
- Sort records alphabetically for better organization.
- Provide a simple console-based or menu-driven user interface.
- Support future extension for intelligent search features.



TOOLS AND TECHNOLOGIES



PROGRAMMING LANGUAGE

- Java

DATA STRUCTURES

- Arrays
- Linked Structures
- Hash Tables

ALGORITHMS USED

- Linear Search
- Binary Search
- KMP Algorithm
- Rabin–Karp Algorithm
- Merge Sort / Quick Sort

DEVELOPMENT ENVIRONMENT

- IntelliJ IDEA
- VS Code
- Eclipse

VERSION CONTROL

- Git & GitHub

EXPECTED OUTCOME



- Reduce the time required to locate library resources.
- Provide fast multi-parameter search functionality.
- Improve search accuracy and retrieval efficiency.
- Maintain an organized digital repository.
- Demonstrate the implementation of DSA concepts in a real-world application.
- Provide a foundation for future enhancements such as fuzzy search and ranked results.

FINAL CONCLUSION

The Digital Library Search System is a Java-based, DSA-oriented application that provides an efficient and scalable approach for storing, searching, and retrieving library resources without relying on a database management system.

THANK YOU

Questions & Discussion

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